

THE UNIVERSITY OF
SYDNEY

Tutorial Class	
Student Name	
SID	

CONFIDENTIAL QUIZ PAPER

This paper should not be removed from the quiz venue.

COMP5339 – Data Engineering

Tutorial Quiz – 1

Individual Quiz (5%)

Quiz Writing Time: **30 minutes**
Friday, 22 May, at 06 PM

Instructions to Students

- Please ensure you correctly complete your **tutorial** and **Student ID** details.
- This is a **closed-book** test. **Electronic devices** (including phones, laptops, tablets, etc.) are **not permitted**. Please place all devices in your bag and keep your bag on the floor.
- This tutorial quiz consists of **3 multiple-choice questions** and three short descriptive questions. **All questions must be answered.**
- Write all answers in the spaces provided on this paper.
- Questions carry **different marks**. The mark value for each question is shown at the start of the question. The **total mark** for this quiz is **10**, which contributes to 5% of the total marks.
- For short-answer questions, please write **legibly**. Record your final answers in **ink**. **Do not** use a pencil or **red ink**.

Please submit the completed quiz paper to your tutor **before leaving the room**.

Part I – Multiple Choice Questions

Question 1: This question has three parts. Tick the boxes corresponding to each correct answer. Each question carries one mark.

A. In a supervised text-classification pipeline, which sequence is most appropriate?

- ☐ **Raw text → feature extraction → labels + features → model training**
- ☐ Raw text → clustering → labels → storage
- ☐ Raw text → image similarity → prediction
- ☐ Raw text → regression → tokenisation

B. In a publish/subscribe system, publishers send messages to:

- ☐ Specific consumers directly
- ☐ Window operators only
- ☐ Relational tables only
- ☐ **Topics managed by a broker**

C. Why might one-hot encoding become problematic in a production pipeline?

- ☐ **It may create very high-dimensional data when there are categories**
- ☐ It cannot represent categories
- ☐ It works only on images
- ☐ It always causes label leakage

Part II – Short Answer Questions

Provide answers in the provided boxes.

Question 1: Why is unstructured data important in data engineering? (2 Marks)

Unstructured data is important because a very large portion of useful business information exists in forms such as text, images, video, email, and social media, rather than in structured tables. From a data engineering perspective, this means pipelines must be designed to ingest, preprocess, extract features from, and analyse these data types before they can support machine learning or analytics.

Question 2: Discuss the main goals of scalability in distributed data systems, including speed-up and scale-up. Why are these difficult to achieve perfectly in practice? (2 Marks)

Speed-up means that adding more resources should reduce processing time for the same amount of data. Scale-up means that if both data size and resources grow together, the system should maintain similar performance. In practice, perfect scalability is difficult because coordination, communication, and synchronisation overhead increase as more nodes are added.

Question 3: Why is simply having multiple PostgreSQL servers usually not equivalent to using a parallel database such as Greenplum for large-scale MADlib workloads? (3 Marks)

Multiple PostgreSQL servers are independent systems, so partitioning, distributed query planning, cross-node joins, merging results, and fault handling must be managed manually. Greenplum, in contrast, is designed as a shared-nothing parallel database with coordinated execution across nodes. Therefore, Greenplum usually provides better scalability and more integrated support for data-parallel analytics.

EXTRA SHEET